

西南财经大学本科考试



课程论文

学年学期：2025-2026-2

课程名称：公司金融（英）

论文题目：Capability Surprise as the Pricing Mechanism of an Early-Commercial-Stage Foundation-Model Lab: How Model-Release and Benchmark Events Drive Price Discovery and Drift (Zhipu AI, 2513.HK)

学生学号：42353012

学生姓名：许哲圣

学 院：特拉华数据科学学院

年级专业：2023 级 金融数学

评语：

得 分： _____

评阅教师签字： _____ 年 月

Capability Surprise as the Pricing Mechanism of an Early-Commercial-Stage Foundation-Model Lab: How Model-Release and Benchmark Events Drive Price Discovery and Drift (Zhipu AI, 2513.HK)

Zhesheng Xu (许哲圣) | Student ID: 42353012 | Corporate Finance —Final Project

June 2026

Abstract

In January 2026 Zhipu AI (Knowledge Atlas Technology, 2513.HK) became the first foundation-model laboratory to list publicly anywhere, on Hong Kong’s Chapter-18C board. By 26 June 2026 its shares had risen about seventeen-fold to a market capitalisation near US\$114bn—roughly 570 times its ~US\$200m of expected 2026 revenue—despite a gross margin that has fallen to near zero and negative book equity. We ask what such a firm is worth and how its price forms. A three-scenario DCF, a reverse DCF, a comparables check and a real-options reading place fundamental value at only HK\$29–84 per share (about 2% of the market price); the price instead embeds a winner-take-all outcome of roughly US\$100bn revenue by 2035. Because the firm has no earnings to be surprised by, we replace the earnings-surprise event study with a *capability-surprise* one (model releases and benchmark jumps). The market prices capability immediately and discriminately—separating Zhipu from peers on model quality, not sector—with preliminary evidence of post-capability-announcement drift. The valuation gap is best read as capability momentum priced as an option, not a simple bubble.

Keywords: foundation models; Chapter 18C; pre-profit valuation; real options; market efficiency; capability surprise; PEAD/PCAD.

1 Introduction

The first half of 2026 produced an event with no precedent in capital-market history: within forty-eight hours, two Chinese “AI tiger” laboratories—Zhipu AI (2513.HK, 8 January) and MiniMax (00100.HK, 9 January)—became the first foundation-model developers to go public anywhere, listing on the Hong Kong Stock Exchange. Zhipu listed under Chapter 18C, the regime for specialist-technology firms that admits pre-profit issuers. By 26 June 2026 its shares had compounded roughly seventeen-fold to a market capitalisation near US\$114bn on FY2026 expected revenue of about US\$200m—an implied price-to-sales multiple near $570\times$ [Tushare, 2026, Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI), 2025].

This paper poses three questions: **(i)** what is Zhipu worth on fundamentals; **(ii)** what does the re-rating imply—efficient pricing of capability, exuberance, or option value; and **(iii)** is the open-weight, cost-disruption strategy sustainable? Our contribution is methodological: because an early-stage lab has no earnings, the standard earnings-surprise event study [Ball and Brown, 1968, Bernard and Thomas, 1989] has nothing to bite on, so we reframe the information event as a *capability surprise*—a model release or benchmark jump—adapting the surprise-event (*chao yuqi*) design of Guosen Securities Economic Research Institute [2020] to a new asset class. The weight of the paper falls on the valuation (Section 4) and the event study (Section 5); Sections 2–3 give only the context they require.

2 Background: Company, Financials, and Listing

Business model. Founded in 2019 out of Tsinghua University’s Knowledge Engineering Group, Zhipu develops the GLM (“General Language Model”) family and monetises through usage-

based API access, enterprise subscriptions, and private/on-premise deployment. Its strategic choice is an *open-weight funnel*: flagship models are released with open weights and aggressive token pricing, converting developer mindshare into a usage flywheel via the Z.ai and Z.code platforms.

Financials (full statements in Appendix A). Three facts dominate. (1) Revenue is explosive but tiny: RMB 312.4m (2024) $\rightarrow$ RMB 724.3m (2025, +132%), i.e. roughly US\$44m then US\$102m. (2) The gross margin has *collapsed* from 76.1% (2022, a normal software profile) to 3.4% (2024) and to -0.4% in H1 2025, as the firm bought share through compute-heavy, under-priced workloads. (3) Losses and leverage are deepening: the 2025 net loss was about RMB 4.70bn (+237%), and at 30 June 2025 the group reported net liabilities of RMB 6.15bn against cash of only RMB 190.9m [Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI), 2025]. Zhipu is therefore *early-commercial-stage and pre-profit*, with negative book equity—so ratio analysis is of limited use and the informative metrics are growth, the gross-margin path, and cash runway.

Listing and market reaction. Zhipu priced at HK\$116.20, raising about HK\$4.3bn with HK\$3bn of cornerstone commitments, at an implied valuation near US\$6.7–7.4bn [Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI), 2025]. The listing is strategically remarkable on three counts: Chapter 18C gave a loss-making lab permanent public capital years before profitability; being the world’s first listed foundation-model company conferred a certification premium; and the stock then re-rated to about HK\$2,046 by 26 June 2026 ($\approx +1,660\%$ versus IPO; realised annualised volatility near 193%; Figure 1). MiniMax, listing one day later at HK\$165, doubled on debut and then *de-rated* by roughly 60% from its peak—a divergence we exploit in Section 5.

3 Industry, Macro, and Strategy

Structure and strategy. In Porter terms the foundation-model layer is harsh: supplier power (advanced compute) is high and politically rationed, buyer switching costs are low because open weights commoditise the layer, and the substitutes are rival open-weight models. Zhipu’s answer is leaderboard dominance through *cost disruption*: a mixture-of-experts GLM-5 line (GLM-4.6: $\approx 355\text{B}$ parameters, 32B active, 200K context) that topped coding benchmarks—GLM-5.1 ranked #1 on SWE-Bench Pro at 58.4%, and GLM-5.2 (June 2026) shipped with MIT open weights and a one-million-token context—priced below closed competitors, now paired with nascent pricing power (8–17% API price rises with each GLM-5.x release) [Zhipu AI / Z.ai, 2026]. This cadence defines the event taxonomy of Section 5. The main weakness is self-inflicted: the same cadence commoditises its own product, while the near-zero gross margin and compute dependence cap monetisation.

The Hong Kong AI-IPO landscape. The 2026 listing wave spans two business models. Zhipu and MiniMax are general-purpose foundation-model laboratories; a third entrant, **Wenge AI** (中科闻歌, 01956.HK), listed on 26 June 2026 at HK\$60.70 and closed its first day near HK\$111.7 (+84%), at about HK\$10.5bn ($\sim$ US\$1.3bn) [Wenge Technology Co., Ltd. (中科闻歌), 2026]. Founded in 2017 by a Chinese Academy of Sciences (Institute of Automation) team, Wenge ranked first by revenue in China’s enterprise decision-intelligence large-model market in 2025 ($\sim 10.2\%$ share), selling a DIOS decision-intelligence operating system and the Decitron and Yayi models mainly to government, media, and enterprise clients. *Recent Hong Kong AI listings include foundation-model laboratories such as Zhipu AI and MiniMax, as well as enterprise-focused AI platforms such as Wenge AI; however, Wenge is better classified as a decision-intelligence solution provider than as a direct frontier-model comparable.* It therefore enters this paper as cross-sectional context, not as a valuation comparable (Section 4.4).

Macro and flows. Chinese AI is a policy priority (state/SOE demand); U.S. compute-export controls force the architectural efficiency that doubles as Zhipu’s cost edge; and Hong Kong’s market structure sets the marginal buyer—Zhipu joined the Hang Seng Tech Index (5 June) and Southbound Stock Connect (8 June 2026), drawing about HK\$920m of net southbound buying on its first eligible day [Hang Seng Indexes Company Limited, 2026]. These flow events matter for Section 5, where we separate them from capability events.

4 Valuation

4.1 Cost of capital

We estimate the cost of equity with the CAPM. Lacking a usable local index series, we relevel a bottom-up beta from global AI-software comparables, giving $\beta \approx 1.6$. With a risk-free rate of 4.0% and an equity risk premium of 6.0%, $K_e = 4.0\% + 1.6 \times 6.0\% = 13.6\%$. As the firm is effectively all-equity, we take WACC $\approx 13.5\%$. Throughout we use HK\$7.8/US\$ and (for RMB statements) RMB 7.1/US\$, and ≈ 435 m shares outstanding.

4.2 Scenario DCF

We project FCFF over 2026–2035 from a US\$200m base, with revenue growth fading from a scenario-specific start to a mature rate, operating margin ramping from -30% to a terminal level, a 15% tax rate, and reinvestment via a sales-to-capital ratio of 1.6–2.0; terminal value uses $g = 4\%$. All cells are live in the companion workbook (`model/valuation_model.xlsx`); the full Base-case chain is in Appendix B, and headline results in Table 1.

Table 1: Three-scenario DCF (USD unless noted; per-share in HKD). Market row at 2026-06-26.

Scenario	Rev. CAGR '26–35	Term. margin	Equity value	Per share (HK\$)
Bear ($p = 0.35$)	21%	18%	US\$1.6B	29
Base ($p = 0.45$)	31%	28%	US\$2.5B	46
Bull ($p = 0.20$)	43%	35%	US\$4.7B	84
Probability-weighted	—	—	US\$2.6B	47
<i>Market (2026-06-26)</i>	—	—	<i>US\$114B</i>	<i>2,046</i>

The probability-weighted intrinsic value is about HK\$47 per share—roughly **2% of the market price**. No reasonable point estimate of fundamentals approaches the quote.

4.3 Reverse DCF: what the price requires

We invert the model to ask what would justify the US\$114bn equity value. Holding WACC at 13.5% and a 40% terminal margin, the price requires sustained revenue growth on the order of 100% per year, implying revenue **on the order of US\$100bn by 2035** (roughly a 100% ten-year CAGR, several hundred times FY2026E). The exact figure is sensitive to the reinvestment assumption, but the qualitative message is robust: the price embeds a hyperscaler-tier outcome reserved for a handful of global winners.

4.4 Relative valuation

On FY2026E revenue Zhipu trades near **570 $\times$ sales**. For context, leading private frontier labs (OpenAI, Anthropic) have transacted around 30–40 $\times$ trailing revenue, and MiniMax—the most appropriate direct comparable—trades far lower than Zhipu on the same basis. We deliberately *exclude* Wenge AI from the multiple comparison: its government/enterprise project-delivery model is not comparable to a frontier model-API and developer-ecosystem business. Zhipu is thus priced roughly an order of magnitude above even the richest private peers—consistent with the reverse-DCF reading rather than with comparable multiples.

4.5 A real-options reading

The reconciliation is that Zhipu is best valued not as a discounted cash-flow stream but as a *call option on a winner-take-all AGI outcome* [Black and Scholes, 1973, Damodaran, 2018]. The DCF base case is the option’s modest intrinsic value; the bulk of the market price is time value on a fat right tail—high volatility ($\approx 193\%$ annualised) and a long horizon both inflate that value. This separates *fundamental value* from *option value* and explains why the stock can rationally trade far above any DCF point estimate while remaining acutely sensitive to capability news—the subject of Section 5.

5 Capability Surprise: An Event Study

Hypothesis. If the firm is priced on its capability trajectory, then capability information should move the price. Under semi-strong efficiency [Fama, 1970] the reaction should be immediate and complete; any systematic *post-capability-announcement drift* (PCAD)—the analogue of the post-earnings drift of Ball and Brown [1968], Bernard and Thomas [1989]—would indicate inefficiency.

Method. On daily prices [Tushare, 2026] we compute mean-adjusted abnormal returns and CARs over three windows—leakage $[-5, -1]$, reaction $[0, +1]$, and drift $[+2, +10]$ —following standard event-study practice [MacKinlay, 1997] and the surprise-event design of Guosen Securities Economic Research Institute [2020]. We first detected Zhipu’s largest abnormal up-days *blind*; they coincide almost one-for-one with GLM releases (Figure 2), validating the event dates against price action.

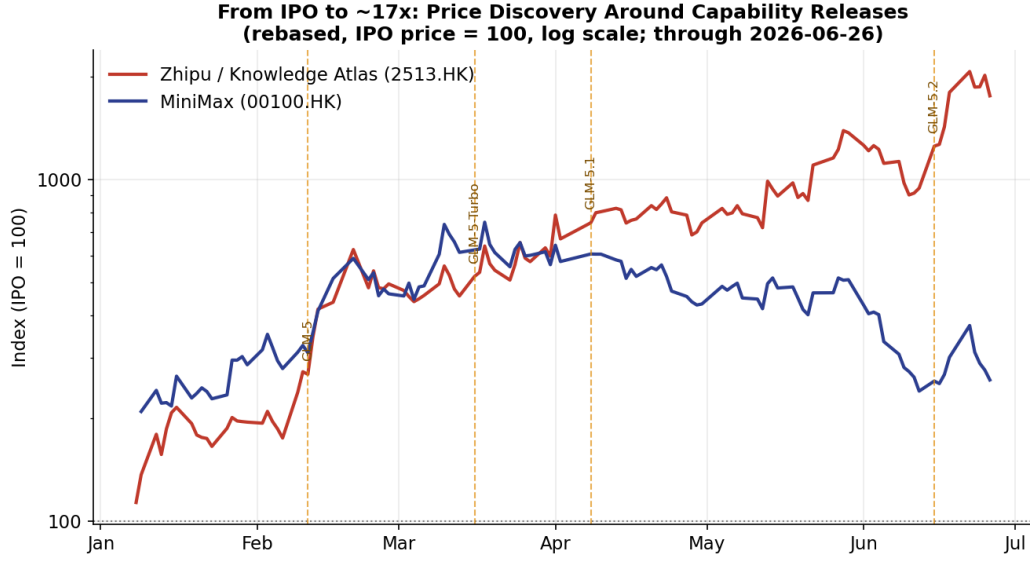


Figure 1: Rebased prices (IPO = 100, log scale, through 2026-06-26) with GLM release markers. Same sector, divergent paths: Zhipu re-rates on successive SOTA releases while MiniMax de-rates from its peak.

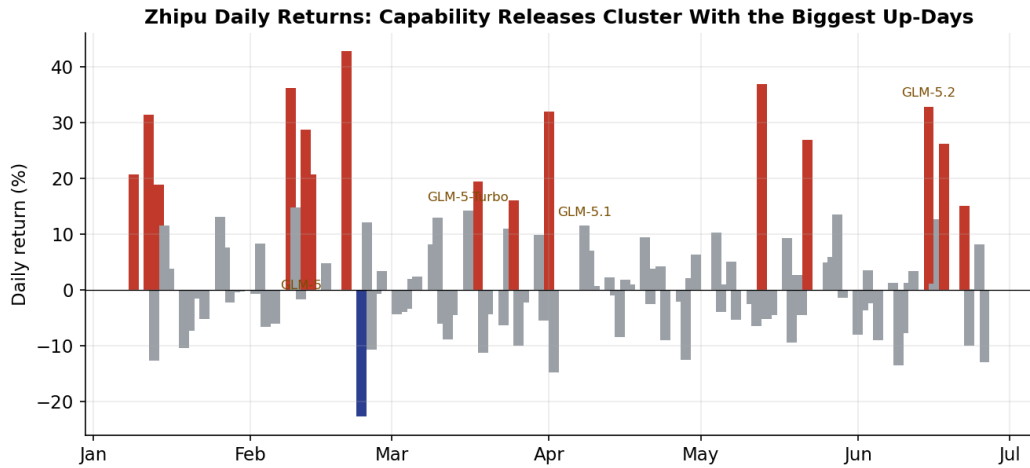


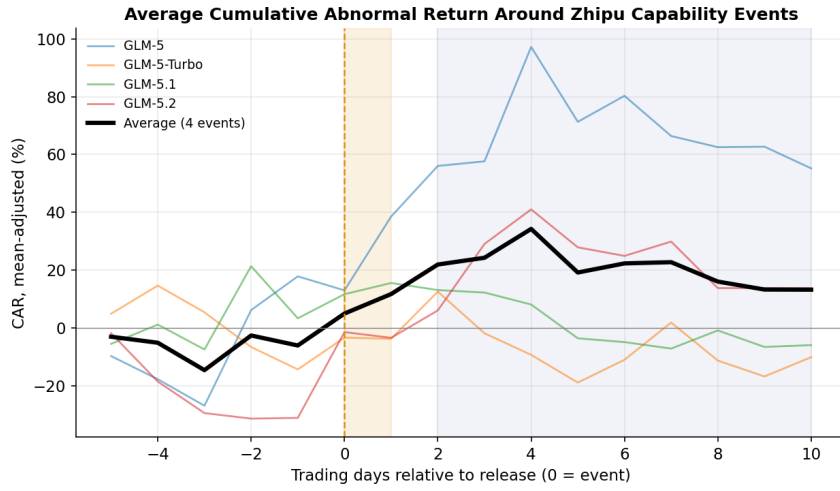
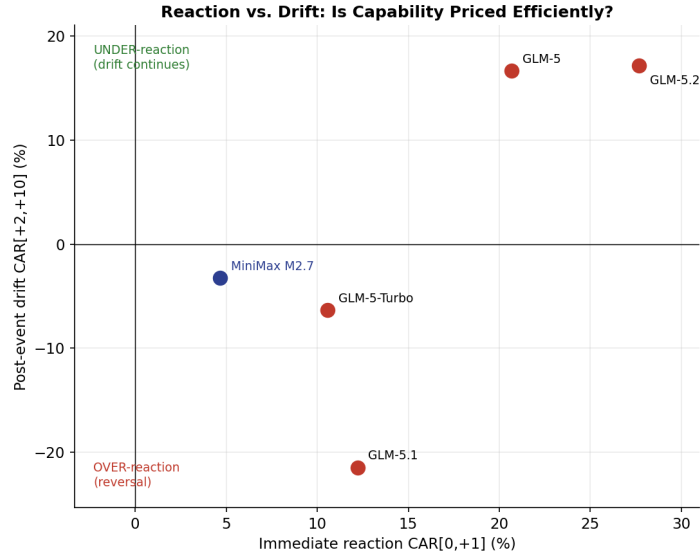
Figure 2: Zhipu daily returns. The largest up-days cluster on capability releases—the market trades a “capability calendar.”

Results. Averaged across the four GLM events, leakage is small, the reaction window is +17.8% (the market prices capability, fast), and the drift averages only +1.5% but is strongly *bimodal* (Table 2). The drift sign depends on the *type* of release: a genuine SOTA leap (GLM-5.2) continues upward, whereas an incremental upgrade (GLM-5.1) reverses. MiniMax’s M2.7 produced a muted reaction and then de-rated, consistent with *capability surprise, not sector membership, driving the cross-section* (Figures 3–4).

Table 2: Cumulative abnormal returns around capability events (mean-adjusted, %).

Event	Day 0	Reaction [0, +1]	Drift [+2, +10]	Reading
GLM-5	2026-02-11	+20.7	+16.7	under-reaction
GLM-5-Turbo	2026-03-16	+10.6	−6.4	mild reversal
GLM-5.1	2026-04-08	+12.2	−21.5	over-reaction
GLM-5.2	2026-06-15	+27.7	+17.2 [†]	strong under-reaction
<i>MiniMax M2.7</i>	2026-03-18	+4.7	−3.3	muted / de-rate
Average (4 GLM)		+17.8	+1.5	

[†] GLM-5.2's [+2, +10] window is *truncated* (only 7 of 9 trading days fall on or before the 2026-06-26 cut-off); its drift is therefore provisional.

**Figure 3:** Average CAR in event time across the four GLM events.**Figure 4:** Reaction versus drift. SOTA leaps sit in the under-reaction quadrant; incremental upgrades in the over-reaction quadrant.

5.1 Not every spike is capability: flow events

Two of the largest blind-detected up-days do *not* map to a model release: +43% on 2026-02-20 and +37% on 2026-05-13. Both coincide with *index/flow* catalysts—rising expectations of Hang Seng Tech inclusion and Stock Connect access (realised 5 and 8 June). Separating capability events from flow events matters: the CARs above are driven by the former, while the latter explain part of the raw volatility and warn against attributing every move to model quality.

5.2 Robustness: peer-adjusted abnormal returns

Because Zhipu and MiniMax share sector and macro shocks, we re-run the study with *peer-adjusted* returns (benchmark = MiniMax’s same-day return). Table 3 shows the reaction result is highly robust (+17.8% vs +18.5%), while the average drift *strengthens* (+1.5% $\rightarrow$ +19.2%): once the falling sector is removed, all four events drift positively, and GLM-5.1’s apparent reversal flips to continuation (−21.5% $\rightarrow$ +12.9%). The under-reaction pattern is reinforced, not weakened, by peer adjustment.

Table 3: Robustness: mean-adjusted vs peer-adjusted CAR (%).

Event	Mean-adjusted		Peer-adjusted	
	React [0, 1]	Drift [2, 10]	React [0, 1]	Drift [2, 10]
GLM-5	+20.7	+16.7	+17.2	+13.3
GLM-5-Turbo	+10.6	−6.4	+14.6	+19.2
GLM-5.1	+12.2	−21.5	+13.4	+12.9
GLM-5.2	+27.7	+17.2	+28.7	+31.3
Average	+17.8	+1.5	+18.5	+19.2

Honesty box. This is *preliminary, diagnostic* evidence consistent with PCAD, not a proven anomaly: $n = 4$ single-firm events (plus one peer) over five months; GLM-5.2’s drift window is truncated; and drift windows can overlap a fast release cadence. No statistical significance is claimed—four events make an interesting undergraduate case, not a general law. The natural extensions are a wider, multi-lab event panel (adding MiniMax and Wenge releases) and an NLP-scored surprise magnitude.

6 Conclusion and Implications

A disciplined valuation cannot reconcile Zhipu’s fundamentals (HK\$29–84 per share) with its market price (HK\$2,046 on 26 June 2026). The reverse DCF shows why: the price embeds a winner-take-all, US\$100bn-revenue outcome, best expressed as a call option on frontier AI rather than a cash-flow stream. The event study shows *how* that price is discovered—through capability surprises priced immediately and discriminately, with preliminary evidence of under-reaction (a PCAD that survives peer adjustment), while separate index/flow events drive part of the rest. The verdict is neither “efficient” nor “bubble” but **capability momentum priced as an option**. For investors, this argues for trading the capability calendar with attention to release *type*; for founders, Chapter 18C has made capability milestones the primary currency of value; for policymakers, compute access is now a direct input to capital formation. The main limitations are the small event sample and a near-zero gross margin whose future path is highly uncertain.

A Financial Statements Summary

From the Chapter 18C prospectus and the H1 2025 interim disclosure (RMB; USD at ≈ 7.1).

Table 4: Income-statement summary (RMB millions unless noted).

	FY2022	FY2023	FY2024	FY2025
Revenue	—	≈ 120	312.4	724.3
growth	—	—	—	+132%
Gross margin	76.1%	—	3.4%	$\approx 0\%$ (H1: -0.4%)
Net loss	—	—	2,958.0	$\approx 4,700$
loss growth	—	—	—	+237%

Table 5: Balance-sheet highlights (RMB millions, at 30 June 2025).

Item	Amount
Cash and cash equivalents	190.9
Net liabilities (negative equity)	$-6,150.8$
Net current liabilities	$-9,564.8$
Shares outstanding (post-IPO, approx.)	$\approx 435\text{m}$
IPO price / gross proceeds	HK\$116.20 / $\approx$ HK\$4.3bn

The gross-margin collapse from 76% to near zero is the central operating fact: Zhipu is converting a high-margin software model into a compute-subsidised, share-grab model. Combined with negative equity, it is why the valuation in Section 4 must rest on a future-option logic rather than on current fundamentals.

B Base-Case DCF Projection

Base-case chain (US\$m unless noted), reproducing the live workbook: WACC = 13.5%, tax = 15%, terminal $g = 4\%$, sales-to-capital = 1.8, ≈ 435 m shares, HK\$7.8/US\$. Revenue grows from US\$200m fading $62\% \rightarrow 8\%$; operating margin ramps $-30\% \rightarrow 28\%$.

Table 6: Base-case projection (US\$ millions unless noted).

Year	Revenue	Growth	EBIT margin	NOPAT	Reinvest	FCFF
2026	200	62%	-30%	-51	43	-94
2027	312	56%	-24%	-62	62	-125
2028	468	50%	-17%	-68	87	-155
2029	674	44%	-11%	-61	114	-176
2030	930	38%	-4%	-33	142	-176
2031	1,228	32%	2%	23	165	-142
2032	1,547	26%	9%	114	177	-63
2033	1,856	20%	15%	238	172	67
2034	2,116	14%	22%	388	144	243
2035	2,285	8%	28%	544	94	450

Terminal value (2035) = $450 \times 1.04 / (0.135 - 0.04) = \text{US\$}4,925\text{m}$; PV of TV = US\$1,388m.

Enterprise value = US\$1,040m; + net cash US\$1,500m = equity US\$2,540m = **HK\$46/share**.

References

- Ray Ball and Philip Brown. An empirical evaluation of accounting income numbers. *Journal of Accounting Research*, 6(2):159–178, 1968.
- Victor L. Bernard and Jacob K. Thomas. Post-earnings-announcement drift: Delayed price response or risk premium? *Journal of Accounting Research*, 27:1–36, 1989.
- Fischer Black and Myron Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.
- Aswath Damodaran. *The Dark Side of Valuation: Valuing Young, Distressed, and Complex Businesses*. Pearson Education, 3rd edition, 2018.
- Eugene F. Fama. Efficient capital markets: A review of theory and empirical work. *Journal of Finance*, 25(2):383–417, 1970.
- Guosen Securities Economic Research Institute. 金融工程专题研究：超预期投资全攻略 [special report on financial engineering: A complete guide to earnings-surprise investing]. Technical report, Guosen Securities, Sep 2020.
- Hang Seng Indexes Company Limited. Index review results and constituent-change announcements. Hang Seng Indexes Co. announcements, Jun 2026. Hang Seng TECH Index inclusion of 2513.HK.
- Knowledge Atlas Technology Joint Stock Company Limited (Zhipu AI). Global offering prospectus (stock code 2513). HKEXnews, <https://www1.hkexnews.hk/listedco/listconews/sehk/2025/1230/2025123000017.pdf>, Dec 2025. Chapter 18C listing document; financials and offering terms.

- A. Craig MacKinlay. Event studies in economics and finance. *Journal of Economic Literature*, 35(1):13–39, 1997.
- Tushare. Tushare pro financial data api: Hong kong daily quotations (`hk_daily`). <https://tushare.pro>, 2026. Daily prices for 2513.HK, 00100.HK, 01956.HK, cross-checked with HKEX.
- Wenge Technology Co., Ltd. (中科闻歌). Global offering prospectus and allotment results (stock code 1956). HKEXnews, Jun 2026. Enterprise decision-intelligence; market-share and offering data.
- Zhipu AI / Z.ai. Glm-4.6 and glm-5 series model cards and technical reports. Hugging Face, <https://huggingface.co/zai-org/GLM-4.6>; Z.ai developer docs, 2026. Architecture (MoE), context length, benchmark results.